

Sample Statistics

AGRON 5130

YOUR NAME HERE

Introduction

The goal of this assignment is to give you hands-on practice working with **sample statistics** and **confidence intervals**, building directly on the concepts introduced in this week's videos. You will work with real agronomic datasets, compute sample-based summary statistics, quantify uncertainty using the standard error, and construct confidence intervals using the t distribution.

Submit a knitted PDF or HTML generated from your R Markdown file to Canvas.

Question 1

1 point

Load the dataset "data/millet_uniformity.csv" and examine the first six rows using the `head()` function.

Pearl millet yield was sampled at 160 locations within a field in Kundewadi, Niphad, in the district of Nasik, India.

```
millet <- read.csv('./data/millet_uniformity.csv')
head(millet)
```

```
##   year row col yield
## 1 1934   1   1  48.4
## 2 1934   2   1  44.8
## 3 1934   3   1  44.4
## 4 1934   4   1  48.8
## 5 1934   5   1  50.4
## 6 1934   6   1  48.4
```

Question 2

1 point

Calculate the **mean**, **standard deviation**, and **number of observations** for millet yield.

```
xbar <- mean(millet$yield)
s <- sd(millet$yield)
n <- length(millet$yield)
xbar
```

```
## [1] 42.015
```

```
s
```

```
## [1] 6.406269
```

```
n
```

```
## [1] 160
```

Question 3

1 point

Calculate the **standard error of the mean** for millet yield.

```
se <- s / sqrt(n)
se
```

```
## [1] 0.50646
```

Question 4

2 points

Calculate the t -value (using the `qt()` function) needed to construct a **95% confidence interval** for the mean yield.

A 95% confidence interval leaves 2.5% of the probability in each tail, so use 0.975. Remember that the degrees of freedom are one less than the number of observations.

```
t.val <- qt(p = c(0.025, 0.975),
            df = n - 1)
t.val
```

```
## [1] -1.974996 1.974996
```

Question 5

2 points

Calculate the **95% confidence interval** for millet yield using the statistics computed above.

```
ci <- xbar + (t.val * se)
ci
```

```
## [1] 41.01474 43.01526
```

Question 6

1 point

Load the dataset "data/safflower_uniformity.csv" and examine the first six rows using the `head()` function.

Safflower yield was sampled 340 times within a safflower field in Eloy, Arizona.

```
saff <- read.csv('./data/safflower_uniformity.csv')
head(saff)
```

```
##   expt row col yield
## 1   E4   1   1   265
## 2   E4   2   1   226
## 3   E4   3   1   181
## 4   E4   4   1   235
## 5   E4   5   1   235
## 6   E4   6   1   282
```

Question 7

1 point

Calculate the **mean**, **standard deviation**, and **number of observations** for safflower yield.

```
xbar <- mean(saff$yield)
s <- sd(saff$yield)
n <- length(saff$yield)
xbar
```

```
## [1] 215.1294
```

```
s
```

```
## [1] 51.73589
```

```
n
```

```
## [1] 340
```

Question 8

2 points

Calculate the **standard error of the mean** for safflower yield.

```
se <- s / sqrt(n)
se
```

```
## [1] 2.805773
```

Question 9

2 points

Calculate the t -value (using the `qt()` function) needed to construct a **90% confidence interval** for the mean safflower yield.

A 90% confidence interval leaves 5% of the probability in each tail, so use 0.95.

```
t.val <- qt(p = c(0.05, 0.95),  
           df = n - 1 )  
t.val
```

```
## [1] -1.649361  1.649361
```

Question 10

2 points

Calculate the **90% confidence interval** for safflower yield using the statistics computed above.

```
ci <- xbar + (t.val * se)  
ci
```

```
## [1] 210.5017 219.7571
```